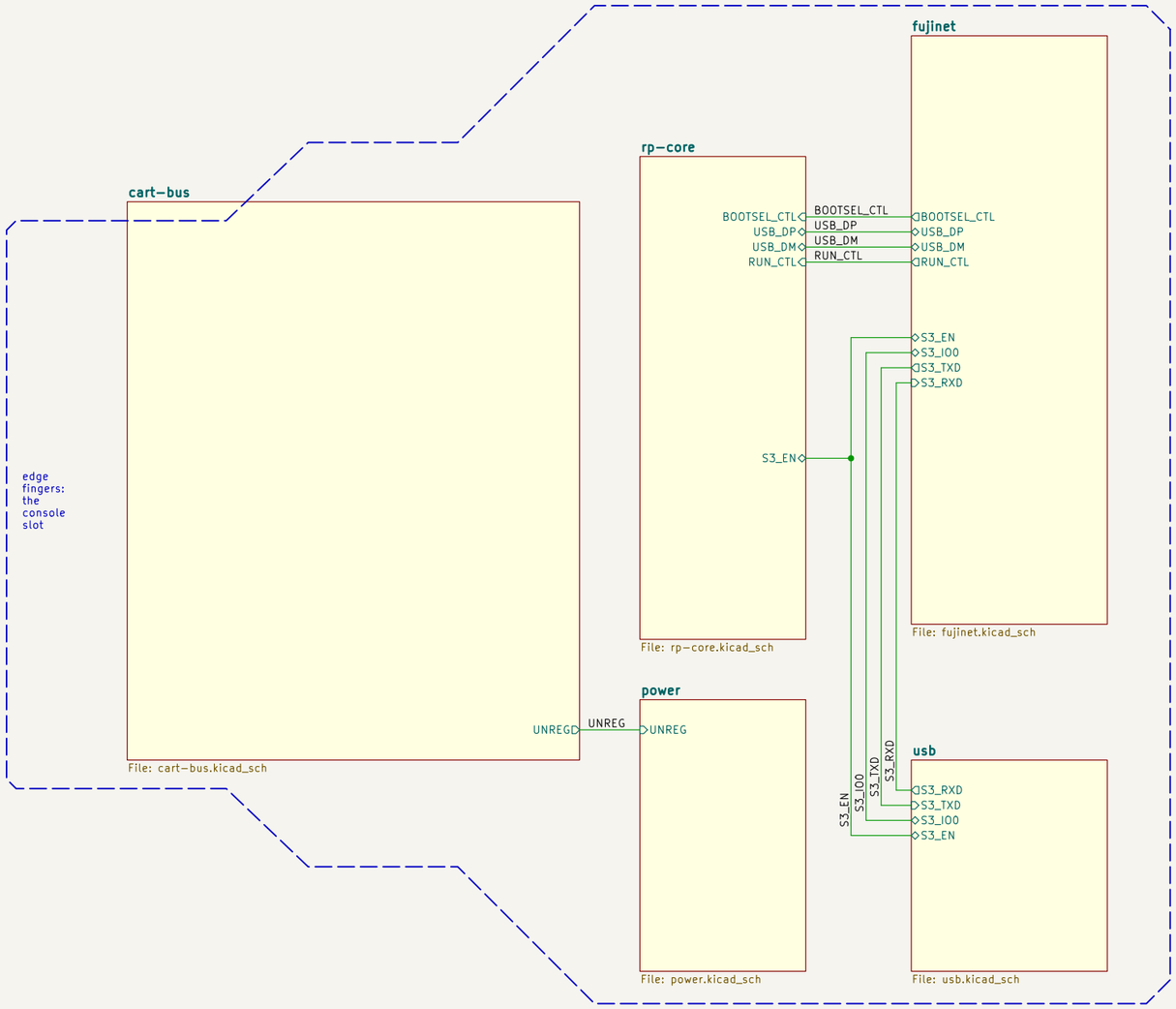


FujiNet for the Atari 5200 SuperSystem, Rev0: RP2354B on the cart bus + ESP32–S3.

This page is the board seen from its component side (the label face), turned so that the edge fingers face left (board south –> page left, north –> right, west –> top, east –> bottom), 2:1, each sheet's block where its circuit sits. Every sheet is drawn the same way: its parts where they are on the board, every connection on it a wire, signals flowing left to right from the console to the USB–C.

cart–bus	the 36–pin edge, the 74HCT541 data buffer, the '08 / '14 enable glue, the RP2354B's 48 GPIOs, the PHI2 Schmitt buffer, the interlock (18 / 36) jumper and its DNP switch, the PWR_OK and ADC senses of the console 5V, the debug UART, the bring–up pads
rp–core	the RP2354B's supplies and decoupling ring, core regulator, crystal, SWD, RUN / BOOTSEL, USB to the S3
fujinet	the ESP32–S3 (FujiNet), microSD, WS2812 status LED, S3 EN / BOOT
usb	USB–C, the CP2102N bridge and esptool's auto–program pair
power	+5V = console 5V (P–FET) OR edge 11 raw DC (5 V buck) OR USB VBUS, the 3.3 V buck (+3V3), the RP's LDO (+3V3_RP)

Between sheets: the nine signals wired here; the rails (GND, CONS_5V, VBUS, +5V, +3V3, +3V3_RP, DVDD) by their power symbols.

The co–design (docs/floorplan–study.md): the 74HCT541 sits over the D0–D7 fingers, its channels taken in reverse so that finger –> '541 –> RP2354B GPIO16–23 is planar on F.Cu (the fan–in is drawn by tools/gen_pcb.py and locked before routing), the RP2354B straight north of it.

Stack–up: 1.6 mm, 6 layers. F.Cu every part, pins 1–18 (the label face), signals + GND pour | In1 GND plane | In2, In3 signals | In4 power: +3V3 north, a +5V island south and east, +3V3_RP under the RP2354B ring, DVDD under its core | B.Cu pins 19–36, signals + GND pour.

Firmware contract (tools/check_nets.py, tools/check_glue.py check this netlist against it): fujinet–firmware pico/atari–5200 a52_cart.h (A0–A13 GPIO0–13, /EN40 GPIO14, /EN80 GPIO15, D0–D7 GPIO16–23, PWR_OK 24, PHI2 25, BUF_EN 26, LED 27, ILK 28, VSENSE ADC GPIO40, debug UART GPIO44 / 45); glue a52_glue_drive(); ESP32–S3 pin map fujiversal–atari5200.h.

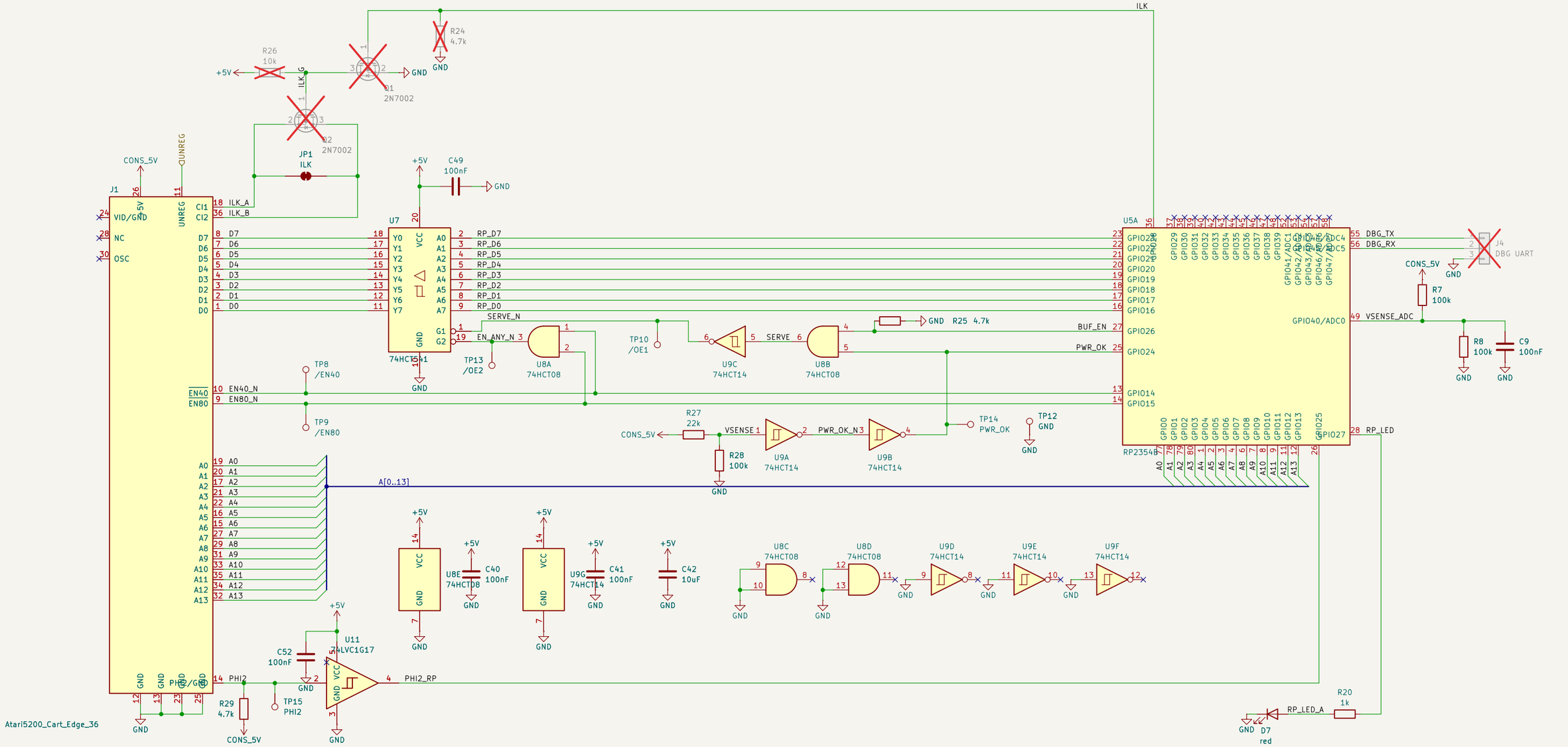
Rev0 2026–10–08: board co–designed with this schematic, see docs/floorplan–study.md
CERN–OHL–W–2.0 (derived from FujiNet–SMS–Rev0 / FujiNet–7800–Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py – edit those, not this file

FujiNet

Sheet: /
File: FujiNet–5200–Rev0.kicad_sch

Title: FujiNet 5200 Rev0

Size: A3	Date: 2026–10–08	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 1/6



Rev0 2026-10-08: board co-designed with this schematic, see docs/floorplan-study.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-7800-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

FujiNet

Sheet: /cart-bus/
File: cart-bus.kicad_sch

Title: FujiNet 5200 Rev0 — Cart bus: edge, RP GPIO, '541 + glue

Size: A3	Date: 2026-10-08	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 2/6

Supplies (RP2350 datasheet 6.3.7): IOVDD, QSPI_IOVDD, USB_OTP_VDD, ADC_AVDD and VREG_VIN on +3V3_RP, the AP2112K LDO that rises with +5V; the core regulator switches VREG_LX through 3.3 uH onto DVDD (VREG_FB senses it), its analog supply VREG_AVDD through 33R / 4.7 uF. One 100 nF at every supply pin, drawn at the pin it serves. RESET resets the RP2354B and the ESP32-S3 together (D5 steers RST_BTN onto RUN and S3_EN). Hold BOOTSEL while pressing RESET for the RP's USB boot ROM. The S3 does both: IO4 (RUN_CTL) low resets the RP, IO5 (BOOTSEL_CTL) low through that reset selects BOOTSEL. QSPI_SD0-3 / SCLK unused: the 2 MB flash is in the package.

S3_EN
D5
BAT54C
RST_BTN
SW1
RESET
GND
+3V3_RP
R21
10k
R14
1k
BOOTSEL_BTN
SW2
BOOTSEL
R15
1k
BOOTSEL_CTL
USB_DP
USB_DM
C24
10uF
C34
100nF
R16
27R
R17
27R
DVDD
L2
3.3uH
R12
33R
PWR_FLAG
C16
4.7uF
C27
4.7uF
TP3
DVDD
C28
100nF
C29
100nF
C25
4.7uF
U5B
QSPI_SS
QSPI_SD1
QSPI_SD2
QSPI_SD0
QSPI_SCLK
QSPI_SD3
QSPI_IOVDD
USB_OTP_VDD
RP_USB_DP
RP_USB_DM
VREG_FB
VREG_VIN
VREG_LX
VREG_AVDD
IOVDD
DVDD
ADC_AVDD
C36
100nF
C15
100nF
C37
100nF
C17
100nF
C18
100nF
C26
100nF
C19
100nF
C20
100nF
RP2354B
XIN
XOUT
SWCLK
SWDIO
RUN
VREG_PGND
GND
C38
100nF
C39
100nF
C14
100nF
R18
1k
Y1
12MHz
XOUT_Y
C30
15pF
C31
15pF
R13
10k
R9
1k
RUN_CTL
TP4
SWCLK
TP5
SWDIO
TP6
GND
TP7
RUN

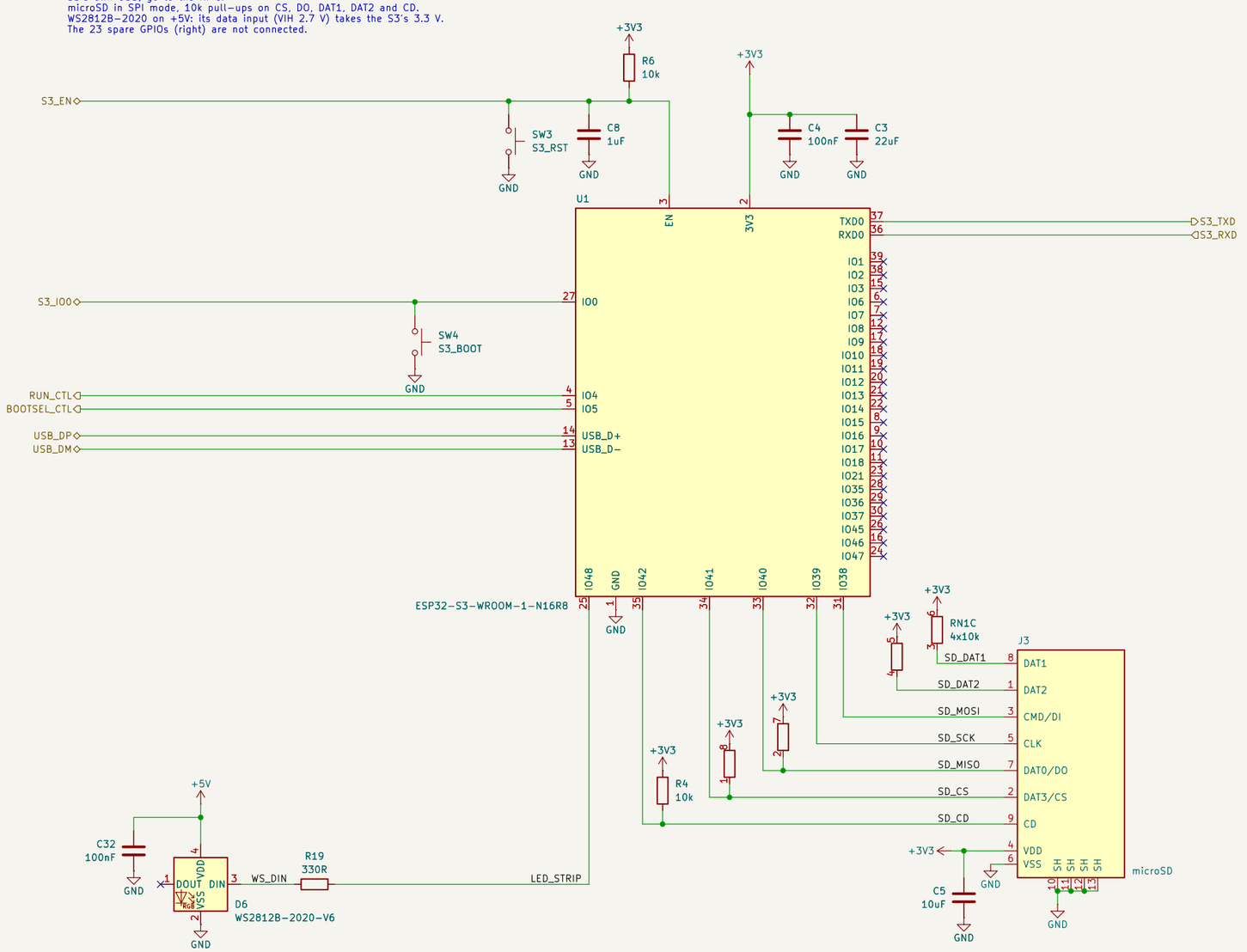
Rev0 2026-10-08: board co-designed with this schematic, see docs/floorplan-study.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-7800-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file
FujiNet

Sheet: /rp-core/
File: rp-core.kicad_sch

Title: FujiNet 5200 Rev0 — RP2354B core: supplies, crystal, USB, RUN

Size: A3	Date: 2026-10-08	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 3/6

ESP32-S3-WROOM-1-N16R8: the FujiNet (fujiversal-atari5200.h pin map).
EN: 10k / 1 uF power-on delay. The S3_RST button, the cart RESET
(rp-core, through the BAT54C) and the USB bridge's auto-program pair
(usb sheet) all pull it low; S3_BOOT and the same pair pull IO0.
IO4 / IO5 drive the RP2354B's RUN and BOOTSEL; IO19 / IO20, the
S3's own USB, go to the RP's.
microSD in SPI mode, 10k pull-ups on CS, DO, DAT1, DAT2 and CD.
WS2812B-2020 on +5V: its data input (VIH 2.7 V) takes the S3's 3.3 V.
The 23 spare GPIOs (right) are not connected.



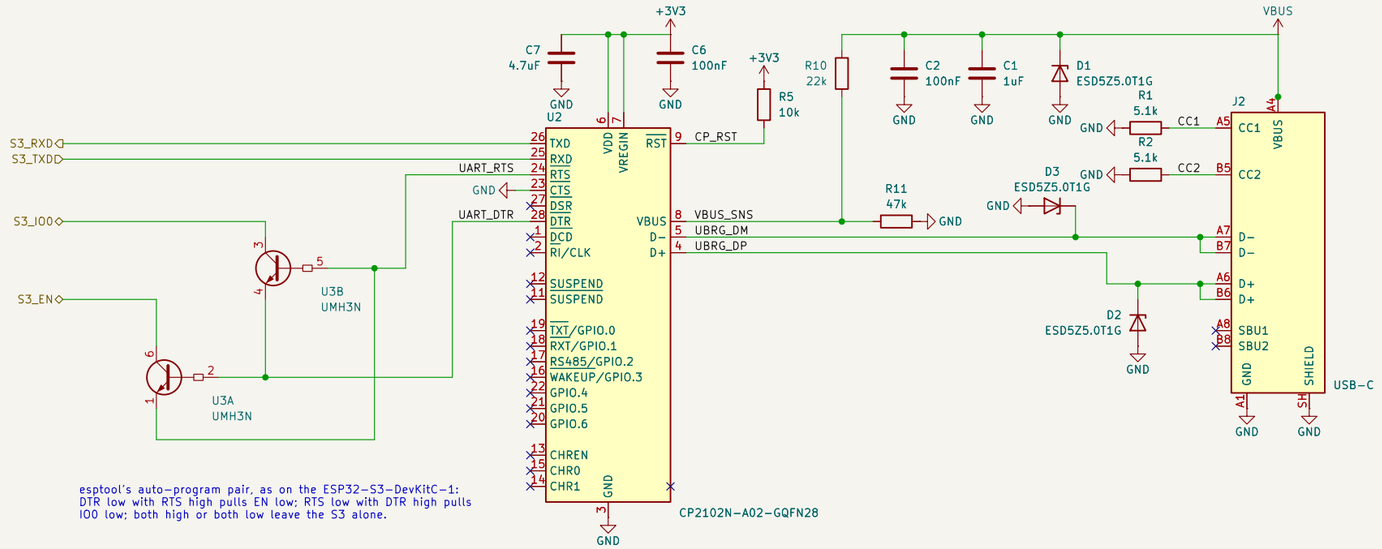
Rev0 2026-10-08: board co-designed with this schematic, see docs/floorplan-study.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-7800-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

FujiNet

Sheet: /fujinet/
File: fujinet.kicad_sch

Title: FujiNet 5200 Rev0 — FujiNet: ESP32-S3, microSD, LED

Size: A3	Date: 2026-10-08	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 4/6



Rev0 2026-10-08: board co-designed with this schematic, see docs/floorplan-study.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-7800-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

FujiNet

Sheet: /usb/
File: usb.kicad_sch

Title: FujiNet 5200 Rev0 — USB-C and the CP2102N bridge

Size: A3	Date: 2026-10-08	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 5/6

Power tree: +5V is the OR of three sources.
  CONS_5V edge 26, the console's VR1 (a 78M05, 0.5 A, on the original CA018087 board, which also feeds the
           console's RAM and controllers), through Q3 AO3401A (D = CONS_5V, S = +5V): on unless its gate CONS_OFF is
           high; with the gate high only its body diode is left, which never passes anything back into the console.
  VBUS   USB-C, through D4 SS34.
  PRE5V   edge 11, the adapter's raw DC (9–13 V; universal boards only, ahead of the power switch, so live whenever
           the adapter is plugged in): JP2 bridged jumper (cut it to take nothing from pin 11), F1 0.35 A PTC, R30 2.2R
           inrush limit, U10 AP63205 5 V buck, D8 SS34. Never ground pin 11: on a universal console it is the adapter.
CONS_OFF = VBUS OR PRE5V (D9 BAT54C, R23 100k): either one turns the console path off, so with raw DC present the
cart takes nothing from VR1. R3 / R22 4.7k hold the SS34s' reverse leakage (0.5 mA at 25 C, 20 mA at 100 C) off
the OR's inputs, so the console path stays on (tools/audit/spice_checks.py). +5V feeds the 74HCT541 / '08 / '14,
the PHI2 buffer, the WS2812, the AP2112K LDO -> +3V3_RP (every RP2354B supply pin: it rises with +5V, so the
5V-tolerant pads are never unpowered with 5 V on them) and the AP63203 buck -> +3V3 (ESP32-S3, microSD, CP2102N).
TP17, TP18, TP16, TP2, TP1, TP11 and the rp-core DVDD pad are bring-up test points.

U6
AP2112K-3.3
TP11
+3V3_RP
C35
1uF
C33
1uF
TP17
CONS_5V
Q3
AO3401A
+5V
C51
10uF
C50
100nF
TP2
C23
22uF
C11
22uF
C12
22uF
C13
100nF
VBUS
D9
BAT54C
CONS_OFF
R23
100k
PWR_FLAG
L3
4.7uH
C43
100nF
C47
22uF
C48
22uF
R22
4.7k
TP16
PRE5V
D8
SS34
JP2
RAWDC
F1
0.35A
UNREG
UNREG_J
UNREG_F
TP18
RAWDC
R30
2.2R
PRE_VIN
PWR_FLAG
U10
AP63205WU
PRE_SW
PRE_BST
PRE5V
C44
22uF
C45
22uF
C46
100nF
TP1
+3V3
L1
6.8uH
C21
22uF
C22
22uF
C10
100nF
BUCK_BST
BUCK_SW
U4
AP63203WU
D4
SS34
R3
4.7k
GND

PWR_FLAG: these rails reach their loads only through passive parts (the edge fingers, USB-C, the P-FET, the SS34s, the bucks' inductors).

PWR_FLAG
CONS_5V PWR_FLAG
+5V PWR_FLAG
VBUS PWR_FLAG
+3V3 PWR_FLAG

Rev0 2026-10-08: board co-designed with this schematic, see docs/floorplan-study.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-7800-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file
FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet 5200 Rev0 — Power: +5V OR, 5V / 3.3V bucks, RP LDO

Size: A3	Date: 2026-10-08	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 6/6